

# Wrapper Types: relational constraints without relational constraints

Guy Emerson

DELPH-IN 2019

# Difference Lists

$$\begin{bmatrix} \textit{diff-list} \\ \text{LIST} \quad \textit{list} \\ \text{LAST} \quad \textit{list} \end{bmatrix}$$

# Difference Lists

data  $\rightarrow$   $\begin{bmatrix} \textit{diff-list} \\ \text{LIST} \quad \textit{list} \\ \text{LAST} \quad \textit{list} \end{bmatrix}$

# Difference Lists

data →  $\begin{bmatrix} \textit{diff-list} \\ \text{LIST} \quad \textit{list} \end{bmatrix}$   
helper →  $\begin{bmatrix} \text{LAST} \quad \textit{list} \end{bmatrix}$

# Wrapper Types

$$\left[ \begin{array}{ll} \textit{wrapper-type} & \\ \text{DATA} & \textit{data-type} \\ \text{HELPER} & \dots \end{array} \right]$$

# Append Lists

$$\left[ \begin{array}{ll} \textit{append-list} & \\ \text{LIST} & \textit{list} \\ \text{APPEND} & \textit{list-of-alists} \end{array} \right]$$

# Append Lists

*closed list* →  $\left[ \begin{array}{l} \textit{append-list} \\ \text{LIST} \quad \textit{list} \\ \text{APPEND} \quad \textit{list-of-alists} \end{array} \right]$

# Append Lists

*closed list* → *append-list*  
*append-lists* → *LIST list*  
to be appended → *APPEND list-of-alists*



# Append-list syntax

```
[ NEW-LIST.APPEND < #1, #2, #3 >,  
  LIST1 #1,  
  LIST2 #2,  
  LIST3 #3 ].
```

# Append-list syntax

```
[ NEW-LIST.APPEND < #1, #2, #3 >,  
  LIST1 #1 & [ LIST < a, b > ],  
  LIST2 #2 & [ LIST < > ],  
  LIST3 #3 & [ LIST < c, d, e > ] ].
```

# Append-list syntax

```
[ SYNSEM...RELS.APPEND < #1, #2, #3 >,
  C-CONT.RELS #1,
  ARGS < [ SYNSEM...RELS #2 ],
          [ SYNSEM...RELS #3 ] > ].
```

# Backwards compatible

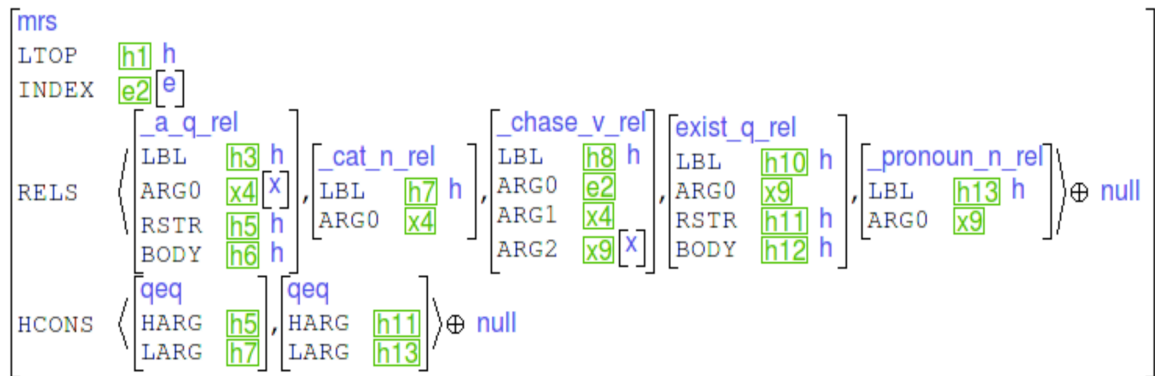
---

- The LIST feature is the same

# Backwards compatible

- The LIST feature is the same
- Refactored Ling-567 English grammar, replacing all diff-lists with append-lists

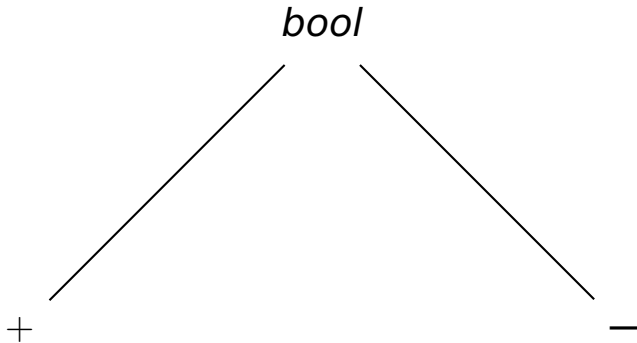
# Backwards compatible



# Computation Types

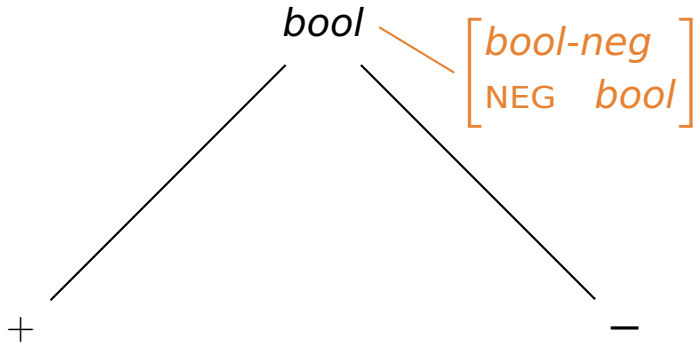
$$\left[ \begin{array}{l} \textit{append-list} \\ \text{LIST} \quad \boxed{1} \textit{ list} \\ \text{APPEND} \quad \left[ \begin{array}{l} \textit{list-of-aliases} \\ \textit{APPEND-RESULT} \quad \boxed{1} \end{array} \right] \end{array} \right]$$

# Computation Types

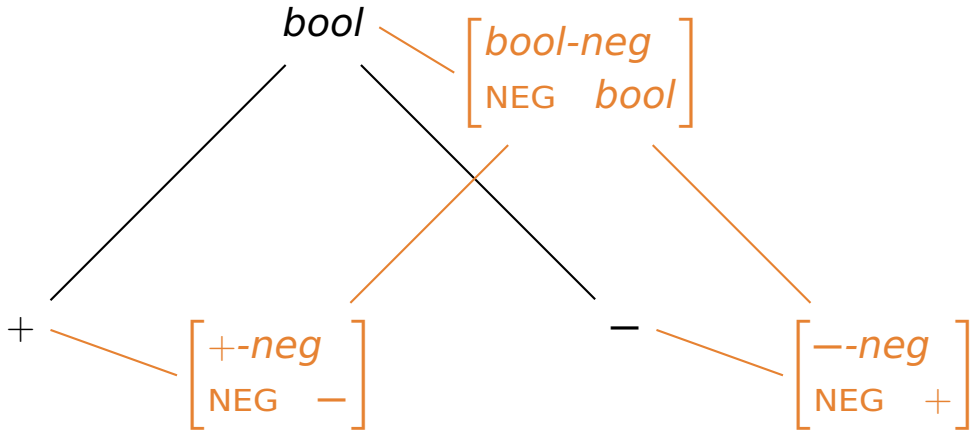




# Computation Types



# Computation Types



# Computation Types

$$\begin{bmatrix} \text{wrapper-type} \\ \text{OUTPUT} \quad \text{data-type} \\ \text{INPUT} \quad \text{computation-type} \end{bmatrix}$$

# Computation Types

$$\left[ \begin{array}{l} \text{wrapper-type} \\ \text{OUTPUT} \quad \boxed{1} \text{ data-type} \\ \text{INPUT} \quad \left[ \begin{array}{l} \text{computation-type} \\ \text{RESULT} \quad \boxed{1} \end{array} \right] \end{array} \right]$$

# Summary of Wrapper Types

- Can manipulate data using computation types
- Very powerful (Turing-complete)
- Easy syntax (e.g. “APPEND < a, b >”)

# Summary of Wrapper Types

- Can manipulate data using computation types
  - Very powerful (Turing-complete)
  - Easy syntax (e.g. “APPEND < a, b >”)
- No need for relational constraints!

# Operations implemented so far

- Lists: append
- Logical: and, or, not
- Arithmetic: add, subtract, multiply, compare